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Stochastic optimization for transportation planning in disaster relief under disruption and uncertainty

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Abstract

Purpose – This study aims to minimize the expected arrival time of relief vehicles to the affected areas, considering the destruction of potential routes and disruptions due to disasters. In relief operations, required relief items in each affected area and disrupted routes are considered as uncertain parameters. Additionally, for a more realistic consideration of the situations, it is assumed that the demand of each affected area could be met by multiple vehicles and distribution centers (DCs) and vehicles have limited capacity.

Design/methodology/approach – The current study developed a two-stage stochastic programming model for the distribution of relief items from DCs to the affected areas. Locating the DCs was the first-stage decisions in the introduced model. The second-stage decisions consisted of routing and scheduling of the vehicles to reach the affected areas.

Findings – In this paper, 7th district of Tehran was selected as a case study to assess the applicability of the model, and related results and different sensitivity analyses were presented as well. By carrying out a simultaneous sensitivity analysis on the capacity of vehicles and the maximum number of DCs that can be opened, optimal values for these parameters were determined, that would help making optimal decisions upon the occurrence of a disaster to decrease total relief time and to maximize the exploitation of available facilities.

Originality/value – The contributions of this paper are as below: presenting an integrated model for the distribution of relief items among affected areas in the response phase of a disaster, using a two-stage stochastic programming approach to cope with route disruptions and uncertain demands for relief items, determining location of the DCs and routing and scheduling of vehicles to relief operations and considering a heterogeneous fleet of capacitated relief vehicles and DCs with limited capacity and fulfilling the demand of each affected area by more than one vehicle to represent more realistic situations.

Keywords Location, Stochastic programming, Relief item distribution, Route disruption, Routing and scheduling

Paper type Research paper

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1. Introduction

Every year man-made and natural disasters such as hurricanes and earthquakes cause significant economic and human losses. In this regard, we can refer to Earthquakes in Iran (2003 and 2017) and Chili (2011 and 2015), Tsunamis in Japan (2011) and the Indian Ocean (2004), floods in the Philippines (2013) and Iran (2019) and Strikes in New York (2005) (Sabouhi *et al.*, 2020; Jabbarzadeh *et al.*, 2018). Therefore, it is vital to make quick and proper decisions in different phases of a disaster because of the unpredictable nature and an increasing number of these disasters. Disaster management is one of the most important issues that is recently being studied in a lot of countries, including Iran. With respect to time factor, disaster management operations can be divided into three categories including before, during, and after the onset of a disaster. Moreover, the following four phases including mitigation, preparation, response and recovery are considered as disaster management phases (Banomyong *et al.*, 2017; Seaberg *et al.*, 2017).

Based on statistics, there is a high rate of fatality that results from a shortage in relief items following a disaster. Therefore, meeting the essential requirements of evacuees, with the purpose of minimizing the arrival time of relief items, would decrease casualties (Safaei *et al.*, 2018). In this regard, transportation and delivery of relief items play vital roles (Ozkapici *et al.*, 2016; Vaez *et al.*, 2020). Relief items consist of the essential requirements of the affected people, such as food, water, tent, clothing, etc. (Berkoune *et al.*, 2012). The demand for these items varies based on the severity of the disaster in each affected area (AA), and is more accurately estimated in the first few hours following a disaster.

Delivery of relief items to the AAs is normally carried out using a fleet of vehicles from several distribution centers (DCs) to the AAs. Any limitation in available resources such as the number and/or the capacity of relief vehicles and DCs, and unexpected accidents like route disruption (destruction of roads, bridges, and tall buildings) could impede the delivery of relief items and consequently delay efficient relief operation in AAs. On the other hand, the number of required items in each AA may exceed the vehicles' capacity. Therefore, it is vital that any AA can be served by multiple vehicles and this characteristic is named split delivery (Moshref-Javadi and Lee, 2016a; Silva *et al.*, 2015). Meeting demand through a vehicle is usually infeasible because of heterogeneity and limited capacity of vehicles. To address these issues, designing an appropriate distribution system by the decision-makers in disaster management field is crucial, and it requires immediate and efficient decisions to minimize risks. These decisions include selecting appropriate locations for DCs among a set of potential locations, allocating vehicles to DCs, and routing and scheduling (R/S) of vehicles.

In recent years, numerous studies have been conducted in the field of relief operations, in the disaster response phase (DRP), but locating DCs and simultaneous R/S of relief vehicles for the supply of relief items to different AAs considering the destruction and disruption of routes and uncertainty, are recognized as new areas of research. This paper contributes to this literature by proposing a two-stage stochastic programming model for transportation planning in disaster relief under disruption and uncertainty. We aim to determine the location of the DCs and R/S of relief vehicles for the supply of relief items among AAs. The goal of the developed model is to minimize the expected arrival time of relief vehicles to the AAs considering split delivery, a heterogeneous fleet of capacitated vehicles and limited capacity for DCs. The application of the presented model is investigated utilizing an actual case problem in 7th district of Tehran.

The remaining part of this research is structured as below: Section 2 reviews the related literature on transportation planning in disaster relief; Section 3 describes the problem statement; Section 4 provides the mathematical modeling of the problem; Section 5 presents the case study of 7th district of Tehran; Section 6 presents the results and sensitivity analyses; and conclusions and future research suggestions are provided in Section 7.

2. Review of the relevant literature

This section aims to review the related literature on location-routing and scheduling (LRS) problem for supply distribution in the DRP.

Many studies in the literature have addressed routing and location problems in relief items distribution (Wang *et al.*, 2014; Rath and Gutjahr, 2014; Vahdani *et al.*, 2018). However, there is limited research on R/S problems in the literature. A multi-objective R/S model for relief items distribution was provided by Hamedi *et al.* (2012). The objectives included reliability and travel costs. Also, they considered multiple depots and split delivery in their model. To solve pickup and delivery problems, Wohlgemuth *et al.* (2012) also presented a multi-period R/S model. Their proposed model aims to increase vehicle utilization and minimize delays. Gan *et al.* (2015) introduced a R/S model for the distribution of relief items from one DC to the AAs, considering utility function. Sabouhi *et al.* (2019) proposed an integrated model that regards R/S of vehicles to transport people from AAs to shelters and provide them with necessary relief items. They assumed that the demand of each AA and DC could be met by more than one vehicle.

To the best of our knowledge, only the models proposed by Moshref-Javadi and Lee (2016b), Wei *et al.* (2020), and Davoodi and Goli (2019) examine simultaneous LRS problems for supply distribution operations. Moshref-Javadi and Lee (2016b) introduced a LRS model which seeks to minimize the total arrival times of the vehicles at AAs. The authors consider multiple depots with heterogeneous fleet of vehicles. Davoodi and Goli (2019) developed a model for locating relief centers and routing vehicles with coverage distance. A bi-objective LRS model was proposed by Wei *et al.* (2020) with two objective functions, consisting of the total cost and the penalty cost caused by time window violations.

In all of the abovementioned studies, the introduced models are deterministic. However, uncertainty of information is the most significant features of a disaster, since many parameters such as demand for relief items in each AA depends on the severity of the disaster and cannot be accurately predicted (Sabouhi *et al.*, 2018; Ghasemi *et al.*, 2019). Hence, a more realistic approach is to consider these parameters as uncertain. One type of uncertainty stems from the random nature of the parameters. Availability of distributional information about random parameters facilitates the application of stochastic programming approaches (Tofighi *et al.*, 2016; Keyvanshokooh *et al.*, 2016). One of the most applicable stochastic programming methods is the two-stage stochastic programming (Fattahi *et al.*, 2017).

In post-disaster relief operations, some works have regarded uncertainty in their model, and mostly applied stochastic programming methods (Tofighi *et al.*, 2016; Galindo and Batta, 2013). For instance, Rawls and Turnquist (2011), Jotshi *et al.* (2009), Mete and Zabinsky (2010), Salmerón and Apte (2010), Rawls and Turnquist (2010), Bozorgi-Amiri *et al.* (2013), Chang *et al.* (2007), Nolz *et al.* (2010), Wang *et al.* (2016), Sheu and Pan (2015), Bozorgi-Amiri *et al.* (2012), Fahimnia *et al.* (2017), Ghasemi *et al.* (2020) and Zhang *et al.* (2019) introduced a model for relief operations in DRP using stochastic programming method. However, routing or scheduling decisions have not been considered in these studies. To the best of our knowledge, in the field of location-routing of relief vehicles in the DRP, the following are studies that have only considered uncertainty and have used stochastic programming method to address it.

Van Hentenryck *et al.* (2010) formulated a multi-depot location-routing model which considered demand and transportation network as uncertain parameters. They used a two-stage stochastic programming method to cope with uncertainty. In their study, the first and second-stage decisions were locating the depots and routing of vehicles, respectively. Rennemo *et al.* (2014) investigated a multi-item location-routing problem for the distribution of relief items from local DCs to the demand points. They considered demand, capacity of vehicles and transportation network as uncertain parameters and presented a three-stage stochastic programming model to solve the problem. First-stage decisions included locating and determining relief items' inventory, whereas second-stage and third-stage decisions were related to distribution of items and routing of vehicles. Caunhye *et al.* (2016) presented a two-stage location-routing model with uncertainty in demand and the state of the infrastructure. Locating depots and determining their inventory level were considered as first-stage decisions, while routing of vehicles was considered as second-stage decision. These studies disregard disruptions and scheduling decisions.

Under disruption and uncertainty, a multi-item location-routing model was introduced by Ahmadi *et al.* (2015) for the distribution of items from DCs to the AAs. Also, the authors developed a two-stage stochastic programming model, regarding the travel time between the nodes as an uncertain parameter due to potential disruptions within the network.

Table 1 illustrates the classification of studies on supply distribution problems with LRS decisions.

Given Table 1 and the foregoing studies on LRS of relief operations in DRP, the research gaps are as follows:

- In the distribution of relief items, few research have conducted a simultaneous analysis of locating, routing, and scheduling decisions, most of which have only focused on locating DCs and routing of relief vehicles. Given the significance of providing relief operations within the shortest possible time, few studies have presented the scheduling of relief vehicles for the fastest delivery to the AAs.
- One of the critical problems that occurs after a disaster is the destruction and disruption of routes used for relief operations, and only few studies on the distribution of relief items has considered this issue.
- Few studies have considered that any AA can be served by more than one vehicle, with multi-depot (multiple DCs) being used for the supply of relief items.
- The severity of a disaster is one of the most influential factors in the decision-making process of disaster management. The demand for relief items in each AA and the number of disrupted routes depends on the severity of the disaster and directly affects the optimal distribution policies. Hence, it is essential to consider these parameters as uncertain and determine their values according to the actual severity of the disaster. However, most of the previous studies have defined the models as deterministic in the field of R/S of relief vehicles.

The contributions of this article are as below:

- introducing an integrated LRS model for the distribution of relief items among AAs;
- applying a two-stage stochastic programming approach to cope with route disruptions and uncertain demands for relief items;
- determining location of the DCs and R/S of relief vehicles;
- considering split delivery, a heterogeneous fleet of capacitated vehicles and limited capacity for DCs to represent more realistic situations; and
- using a real case study in 7th district of Tehran to examine the application of the presented model.

Table 1. Summary of relevant literature

	Decisions			Route conditions			Vehicle fleet				
Authors	location	scheduling	routing	open route	multi-depot	split delivery	heterogeneous	homogeneous	Time window	Uncertainty	Disruption
Van Hentenryck *et al.* (2010)	*		*		*			*		*	
Hamedi *et al.* (2012)		*	*	*		*		*	*		
Wohlgemuth *et al.* (2012)		*	*					*	*		
Rennemo *et al.* (2014)	*		*		*		*			*	
Gan *et al.* (2015)		*	*					*	*		
Ahmadi *et al.* (2015)	*		*		*			*		*	*
Caunhye *et al.* (2016)	*		*		*			*		*	
Moshref-Javadi and Lee (2016a, 2016b)	*	*	*	*			*				
Davoodi and Goli (2019)	*	*	*		*			*			
Sabouhi *et al.* (2019)		*	*	*		*		*			
Wei *et al.* (2020)	*	*	*		*			*	*		
This study	*	*	*	*	*	*	*		*	*	*

3. Problem statement

The occurrence of disasters is a serious threat that causes catastrophic damages and imposes financial losses to a society. In recent years, the increase in casualties and financial losses caused by different disasters has urged societies to be involved in finding effective solutions to minimize the effects of natural disasters such as earthquake, storms, floods, etc.

An influential factor associated with the occurrence of a disaster is selecting appropriate locations for relief facilities among a set of potential locations and performing relief operations via undisrupted routes, and in a timely manner (Golabi *et al.*, 2017). Locating DCs and R/S of vehicles for distribution of items among the AAs, are critical actions that should be taken within the DRP. In the present problem, the supply of relief items from DCs to different AAs is demonstrated in Figure 1. Following the selection of optimal locations for the DCs and allocation of vehicles to these centers, each relief vehicle – such as a truck – will start its route from the assigned DC. The relief vehicle selects the optimal route based on the vehicle's capacity, the amount of required relief items in different AAs, and the undisrupted routes, and will return to its starting point upon completion of the operation.

Route disruptions following the occurrence of a disaster significantly depend on the severity of the disaster. These disruptions urge the selection of the shortest undisrupted routes to provide relief operations in a timely manner. To address the issue of route disruption, independent scenarios with determined occurrence probabilities are considered. Under each scenario, a number of transportation routes are disrupted such that the relief vehicles are unable to access these routes. Furthermore, for each AA under each scenario, a different demand for relief items is considered.

Given the introduced network, the number and location of the DCs and the AAs are determined. The considered relief vehicles are heterogeneous with limited capacity, and each vehicle is assigned to only one DC. The capacity of DCs is also considered as limited. Moreover, the possibility of not using a specific vehicle is considered, and it is assumed that in each scenario, the demand of each AA could be fulfilled by more than one vehicle.

4. Mathematical modeling

In this section, following the introduction of sets, parameters and decision variables, the mathematical model for the problem earlier described will be introduced.

Sets

D = Set of DCs

A = Set of vehicles indexed by $a \in A$

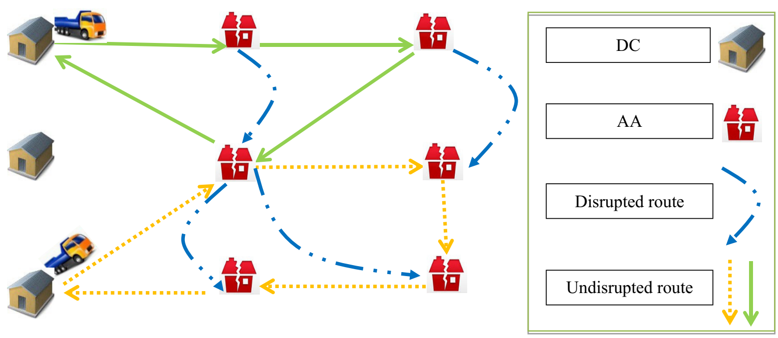

Figure 1.
Relief commodities distribution operation

E = Set of AAs
O = Set of all nodes ($D \cup$E) indexed by $m, k \in N$
S = Set of scenarios indexed by $s \in S$

Parameters

d_{ms} = Demand of AA $m \in E$ under scenario $s \in S$
c_{mk} = Shipment time from node $m \in O$ to node $k \in O$
cap_a = Capacity of vehicle $a \in A$
M_{big} = Sufficiently large number
t_m = Unloading time of relief items in node $m \in E$
p_s = Occurrence probability of scenario $s \in S$
b_{mks} = Equals 1 if a route between node $m \in O$ and node $k \in O$ is disrupted under scenario $s \in S$, otherwise, it equals 0
g = Maximum number of DCs
h_m = Capacity of DC $m \in D$

Decision variables

Q_m = Equals 1 if DC $m \in D$ is opened, otherwise, it equals 0.
X_{amks} = Equals 1 if vehicle $a \in A$ travels from node $m \in O$ to node $k \in O$ under scenario $s \in S$, *otherwise*, it equals 0.
Z_{as} = Equals 1 if vehicle $a \in A$ is dispatched under scenario $s \in S$, otherwise, it equals 0.
Y_{ams} = Equals 1 if vehicle $a \in A$ is assigned to AA $m \in E$ under scenario $s \in S$, otherwise, it equals 0.
R_{ams} = Number of relief items that vehicle $a \in A$ loads from DC $m \in D$ or unloads in AA $m \in E$ under scenario $s \in S$
T_{ams} = Arrival time of vehicle $a \in A$ to the node $m \in O$ under scenario $s \in S$

We use a two-stage stochastic programming approach to formulate the problem. In this approach, some of the key decisions must be made prior to the realization of disruption scenarios; these are known as first-stage decisions. The second-stage decisions are made once a disruption scenario is realized (Shapiro *et al.*, 2009). In our model, locating the DCs is the first-stage decisions and R/S of relief vehicles to reach the AAs are recognized as the second-stage decisions. A more detailed description of stochastic programming method and its applications can be found in Birge and Louveaux (2011).

The developed model is as below:

$$Min \sum_{s \in S} \sum_{a \in A} \sum_{m \in E} p_s \, T_{ams} \tag{1}$$

$$\sum_{a \in A} R_{ams} \leq h_m Q_m \qquad \forall m \in D, \forall s \in S \tag{2}$$

$$\sum_{m \in D} Q_m \leq g \tag{3}$$

K

$$\sum_{m\in D}\sum_{k\in E} X_{amks} = Z_{as} \qquad \forall a \in A,\ \forall s \in S \tag{4}$$

$$\sum_{m\in E} X_{amks} = \sum_{m\in E} X_{akms} \qquad \forall k \in D,\ \forall a \in A,\ \forall s \in S \tag{5}$$

$$\sum_{m\neq k, m\in D\cup E} X_{amks} = \sum_{m\neq k, m\in D\cup E} X_{akms} \qquad \forall k \in E,\ \forall a \in A,\ \forall s \in S \tag{6}$$

$$\sum_{m\neq k, k\in D\cup E} X_{akms} = Y_{ams} \qquad \forall m \in E,\ \forall a \in A,\ \forall s \in S \tag{7}$$

$$\sum_{k\in E, m\neq k} X_{akms} = Y_{ams} \qquad \forall a \in A,\ \forall m \in D,\ \forall s \in S \tag{8}$$

$$cap_a\, Y_{ams} \geq R_{ams} \qquad \forall a \in A,\ \forall m \in O,\ \forall s \in S \tag{9}$$

$$\sum_{a\in A} R_{ams} = d_{ms} \qquad \forall m \in E,\ \forall s \in S \tag{10}$$

$$\sum_{m\in E} R_{ams} \leq cap_a\, Z_{as} \qquad \forall a \in A,\ \forall s \in S \tag{11}$$

$$\sum_{m\in D} R_{ams} = \sum_{m\in E} R_{ams} \qquad \forall a \in A,\ \forall s \in S \tag{12}$$

$$\sum_{a\in A} Z_{as} \leq |A| \qquad \forall s \in S \tag{13}$$

$$X_{amks} \leq (1 - b_{mks}) \qquad \begin{array}{c}\forall s \in S,\ \forall a \in A, \forall k \in O\\ \forall m \in O\end{array} \tag{14}$$

$$T_{ams} = 0 \qquad \forall m \in D,\ \forall a \in A, \forall s \in S \tag{15}$$

$$(T_{ams} + t_m + c_{mk}) - M_{big}(1 - X_{amks}) \leq T_{aks} \qquad \begin{array}{c}\forall m \in D \cup E,\ \forall a \in A, \forall k \in E\\ \forall s \in S\end{array} \tag{16}$$

$$Q_m \in \{0, 1\} \qquad \forall m \in D \tag{17}$$

$$X_{amks},\ Y_{ams}, Z_{as} \in \{0,1\} \qquad \begin{array}{c}\forall s \in S,\ \forall a \in A, \forall k \in O \\ \forall m \in O\end{array} \tag{18}$$

$$R_{ams} \geq 0 \qquad \forall m \in E, \forall a \in A, \forall s \in S \tag{19}$$

$$T_{ams} \geq 0 \qquad \forall m \in O, \forall a \in A, \forall s \in S \tag{20}$$

Objective function (1) minimizes the expected value of arrival time of relief vehicles to AAs. Constraint (2) guarantees that a DC should be opened so that vehicles can be assigned to it, and shows the maximum capacity of each DC. Constraint (3) presents the maximum number of DCs that can be opened. Constraint (4) indicates that when a vehicle is dispatched, it leaves a DC towards one of the AAs. Constraint (5) indicates that a dispatched relief vehicle returns to the DC upon completing the relief operation. Constraint (6) shows the flow balance in AAs, such that any vehicle that enters an AA should leave that area upon completion of its operation.

Constraints (7) and (8) represent the relation between X_{amks} and Y_{ams} variables in each AA and DC, respectively. The relation between variables R_{ams} and Y_{ams} is presented by Constraint (9). Constraint (10) indicates the supply of required relief items for each AA. The maximum capacity of each relief vehicle is controlled by Constraint (11). Constraint (12) warrants that the number of relief items loaded from a DC by a vehicle equals the number of unloaded items in different AAs by the same vehicle. Constraint (13) represents the maximum number of available relief vehicles. Constraint (14) indicates that relief vehicles cannot travel through disrupted routes. Travel starting time of each vehicle leaving a DC is set to zero by Constraint (15). Constraint (16) indicates the arrival time of a relief vehicle to an AA. Finally, Constraints (17) to (20) indicate the type of variables.

5. Case study

Tehran is the most populated city of Iran. It has a population of approximately 12 million, and majority of economic, social, and political centers are located therein (Jabbarzadeh *et al.*, 2014). Every year, a large number of disasters such as flood and earthquake in this city cause financial and human losses (Bozorgi-Amiri *et al.*, 2013; Hasani and Mokhtari, 2019). Being covered with several plateaus, Tehran is highly vulnerable to earthquake disasters.

According to Boostan *et al.* (2015), the three major faults in the city of Tehran are as follows:

(1) Masha fault: one of the main faults of central Alborz, located in north of Tehran with a length of 4 km.

(2) Tehran's north fault: located on the hillside of Alborz mountain chain with a length of 35 km.

(3) Faults on north and south side of Rey: the most dominant faults are located on Tehran's southern fields.

Tehran has 22 districts, among which 7th district contains many important organizations and centers. This district consists of 5 zones, and based on the latest statistics, its population is 309,531. Different social classes live within this district, and many construction projects are ongoing.

Because of the significance of disaster threats in this district, a case problem was conducted in this district to design relief operations to tackle the occurrence of an earthquake. An earthquake severely affects the population of an area, and the number of people requiring relief items in each area should be determined based on the geography of that specific area and the decision of disaster management experts. The objective of the present case study includes locating relief items DCs among a set of potential locations and R/S of relief vehicles for an efficient response upon the occurrence of earthquake within this district.

The main resource of data for this case study is based on the decisions of 7th district disaster management experts, and by International Collaboration Agency of Japan. These data include the number of potential DC locations and their capacities; number and capacity of available relief vehicles, and travel time between network nodes and within routes with high probability of destruction and disruption upon the occurrence of an earthquake.

According to recent analyses of earthquakes in Iran, the destruction of buildings is the main cause of casualties. In this paper, the centers of 5 zones of 7th district are considered as AAs (as shown in Table 2). The demand of each AA for relief items depends on the severity of the disaster. The disruption scenarios are classified according to three scales: small, medium and large disruptions (each disruption cluster includes a number of related scenarios). Moreover, disaster management experts can estimate the number of affected people in each AA based on each scenario. Following a disaster, the evacuees would need relief items such as water, food, blankets, tents, etc. The present study considers one pack of relief items for each evacuee.

To distribute relief items among the AAs, locations are required where relief vehicles can load relief items and distribute them through proper routes within the shortest possible time. To meet this requirement, the present study considered five potential DCs among which a maximum of two districts were to be selected. An important fact to note is that these locations are not established but rather pre-constructed buildings that change their function upon selection. Ghasr crisis warehouses, Imam Khomeini Mosalla, Shiroudi Stadium, Andisheh Park, and Valiasr Site were considered as potential DCs. The capacity of each DC was estimated to be 2,500 items. The location of AAs and potential DCs are demonstrated in Figure 2.

For the distribution of relief items, 7 trucks with a capacity of 700 items were considered, and the required unloading time at each AA was considered as 20 minutes. Travel time between all nodes was calculated using Google Earth 7.1.2.20 19, with a consideration of traffic flow. Additionally, as mentioned before, route disruptions such as destruction of routes, bridges, and tall buildings are inevitable consequences of a disaster, and the severity of the disaster determines the number of disrupted routes. Hence, to address the foregoing issue and have a more realistic representation of the situations, routes with a high probability of disruption and destruction are listed in Table 3. These routes are determined by considering the traffic, based on the opinions of the disaster management experts.

Table 2.
Center of zones in 7th district of Tehran

Zones	Affected areas (zones centers)
1	Mahmoudi Street
2	Sarbaz Street
3	Haleh Street
4	Kooshesh Street
5	Norani Street

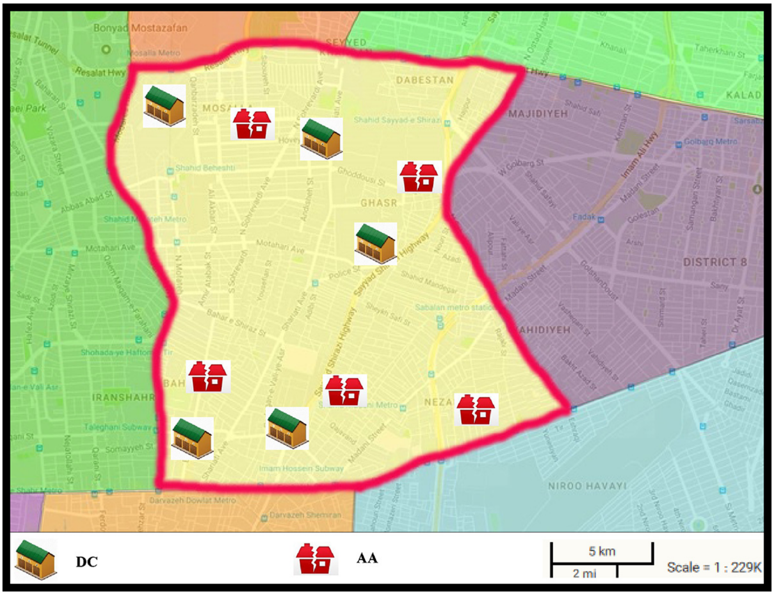

Figure 2.
Location of AAs and potential DCs in 7th district of Tehran

Table 3.
Disrupted routes for different disruption clusters

Disruption Scale	Disrupted routes
Small	Sarbaz Street- Haleh Street/ Mahmoudi Street- Sarbaz Street/ Haleh Street- Shiroudi Stadium
Medium	Sarbaz Street- Haleh Street/ Mahmoudi Street- Sarbaz Street/ Haleh Street- Shiroudi Stadium/ Sarbaz Street- Norani Street/ Kooshesh Street- Norani Street/ Haleh Street- Sarbaz Street -
Large	Sarbaz Street- Haleh Street/ Mahmoudi Street- Sarbaz Street/ Haleh Street- Shiroudi Stadium/ Sarbaz Street- Norani Street/ Kooshesh Street- Norani Street/ Haleh Street- Sarbaz Street/ Kooshesh Street- Haleh Street/ Mahmoudi Street- Norani Street/ Sarbaz Street- Mahmoudi Street

6. Results

The developed model was executed utilizing GAMS 23.0.2 software with CPLEX solver on a computer with Intel Core i7 4702MQ 2.20 GHz up to 3.20 GHz and 6GB RAMDDR3 under Windows® 7. The run time of the model on the case study was 161.84 seconds, with the objective function value of 148.5 time units. Among the 5 potential locations for DCs, Shiroudi Stadium and Andisheh Park were selected. The route of relief vehicles under different disruption clusters is demonstrated in Figures 3, 4, and 5. Furthermore, the number of relief items loaded in each DC and unloaded in each AA, as well as the arrival time of relief vehicles at each of the AAs for different disruption clusters are presented in Table 4.

As an example from Table 3, in large-scale disruptions, a relief vehicle commenced its route from the DC of Shiroodi stadium; arrived at the AA on Haleh St. after 8 time units; unloaded the 399 required relief items of this area; then arrived at the AA on Mahmoudi St. after 39 time units to unload 301 items, and finally returned to Shiroodi stadium DC after 74 time units. Also, this table shows that because of the split delivery assumption, there exist instances that multiple vehicles visit the same AAs to unload relief items.

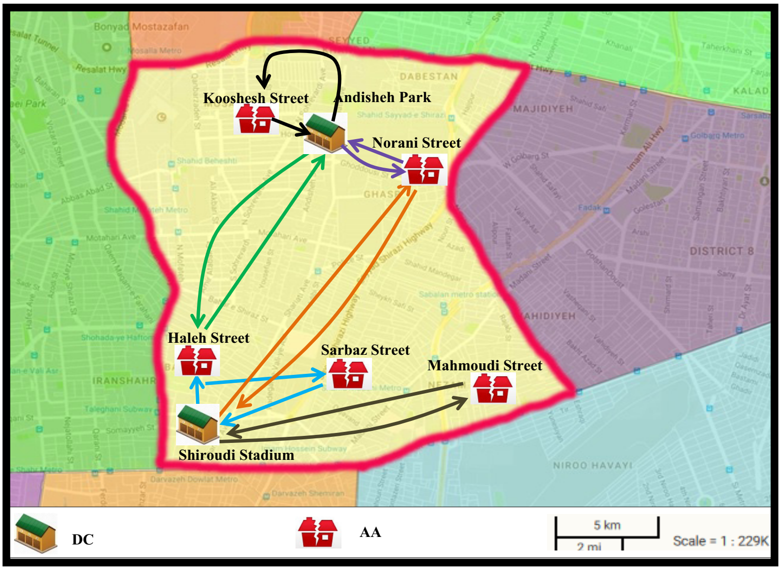

Figure 3.
Route of relief vehicles under small-scale disruptions

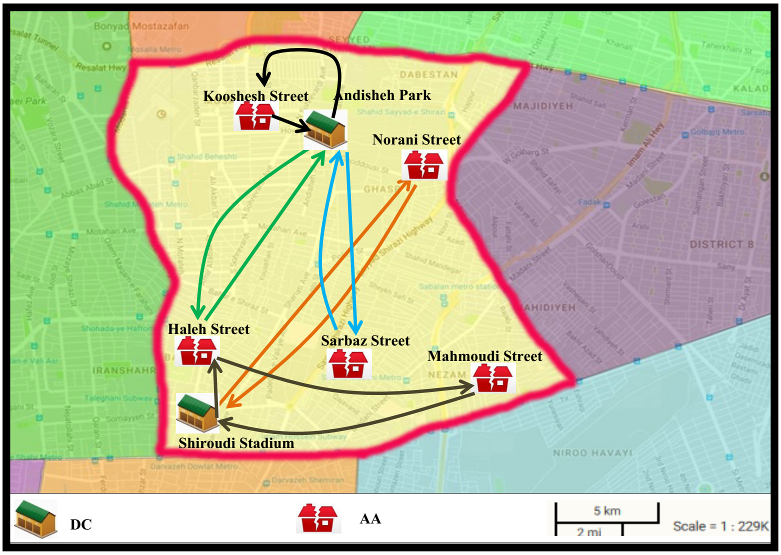

Figure 4.
Route of relief vehicles under medium-scale disruptions

Figure 6 indicates the relief time for situations in which all disruption clusters (stochastic programming), only large-scale disruptions, only medium-scale disruptions, and only small-scale disruptions are accounted for. According to this figure, it is obvious that small-scale and large-scale disruptions provide the lowest and most relief time, respectively. Moreover, total relief operation time under medium-scale disruptions and all clusters disruptions is between those of small-scale and large-scale disruptions. Furthermore, this time in the small-scale disruptions is less than the stochastic programming (all disruption clusters). The

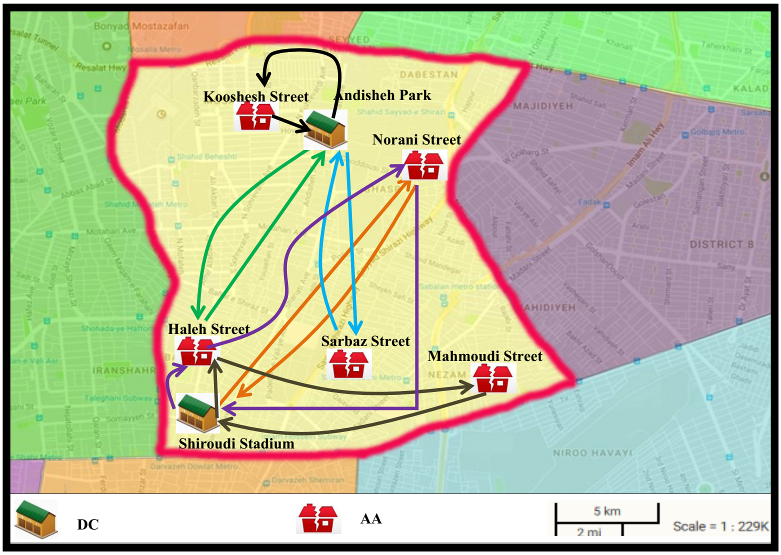

Figure 5. Route of relief vehicles under large-scale disruptions

reason behind this result is that stochastic programming method considers an expected value of all disruption clusters.

Concentrating on the scales of disruption can provide more findings considering the relief time. In small-scale disruptions (earthquake with a low intensity), the demand for relief items in all AAs is minimal, and a lower number of routes are disrupted. This would result in a higher efficiency of distribution process through undisrupted routes, which minimizes the operation time as well. However, the reverse is the case with the large-scale disruptions within which the intensity of the earthquake is higher; more relief items are demanded, and the higher number of disrupted routes increases the delay in relief operation.

The described information in Figure 6 is used for calculating an index known as expected value perfect information (*EVPI*), which is one of the most important indices in stochastic programming (Birge and Louveaux, 2011). This index shows the advantages of considering uncertainty in stochastic parameters. In other words, this index provides the maximum value that a decision-maker agrees to spend on receiving accurate information. *EVPI* is calculated by subtracting the average of objective functions under each independent scenario from the two-stage stochastic model objective function value. *EVPI* of the present study is represented in equation (21):

$$EVPI = 0.1{*}33.9 + 0.4{*}139.2 + 0.5{*}270 - 146.9 = 45.57 \tag{21}$$

Also, we compare the performance of the developed two-stage stochastic programming with an expected value approach. In this regard, an index known value of the stochastic solution (*VSS*) is utilized (Birge, 1982). We consider *EEV* and *RP* as the objective values under expected value approach and two-stage stochastic programming, respectively. Therefore, VSS is written as follows:

$$VSS = EEV - RP \tag{22}$$

RP shows the optimal value of objective function (1) subject to constraints (2)–(20). To compute *EEV*, we gain the expected values of random parameters and optimize objective

Table 4.
Results of commodity distribution operations for different disruption clusters

Disruption Scale	Number of vehicles in each route	Route of each vehicle	Number of commodities loaded in each DC and unloaded in each affected area (commodity)	Arrival time of each vehicle to each node (time unit)
Small	1	Andisheh Park- Haleh Street- Andisheh Park	626–626-0	0–16-52
	1	Shiroudi Stadium- Haleh Street- Sarbaz Street- Shiroudi Stadium	700–364-236–0	0–8-33–68
	1	Shiroudi Stadium- Norani Street- Shiroudi Stadium	700–700-0	0–13-44
	1	Andisheh Park- Norani Street - Andisheh Park	285–285-0	0–12-44
	1	Shiroudi Stadium- Mahmoudi Street- Shiroudi Stadium	103–103-0	0–15-50
	2	Andisheh Park- Kooshesh Street - Andisheh Park	700–700-0 324–324-0	0–8-36
Medium	1	Shiroudi Stadium- Haleh Street-Mahmoudi Street - Shiroudi Stadium	700–498-202–0	0–8-39–74
	1	Andisheh Park- Haleh Street- Andisheh Park	591–591-0	0–16-52
	2	Andisheh Park- Kooshesh Street - Andisheh Park	700–700-0 425–425-0	0–8-36
	2	Shiroudi Stadium- Norani Street- Shiroudi Stadium	700–700-0 385–385-0	0–13-46
	1	Andisheh Park- Sarbaz Street- Andisheh Park	440–440-0	0–13-46
Large	1	Shiroudi Stadium- Haleh Street-Mahmoudi Street - Shiroudi Stadium	700–399-301–0	0–8-39–74
	2	Andisheh Park- Kooshesh Street - Andisheh Park	700–700-0 522–522-0	0–8-36
	1	Andisheh Park- Haleh Street- Andisheh Park	571–571-0	0–16-52
	1	Andisheh Park- Sarbaz Street- Andisheh Park	535–535-0	0–13-46
	1	Shiroudi Stadium- Norani Street- Shiroudi Stadium	700–700-0	0–13-46
	1	Shiroudi Stadium- Haleh Street- Norani Street - Shiroudi Stadium	700–219-481–0	0–8-37–70

function (1) under constraints (2)–(20). For the expected value approach, optimal value of the first-stage decision (i.e. Q_m) is then determined. Finally, we minimize objective function (1) subject to constraints (2)–(20), while fixing the value of the first-stage decision equals to the value gained in the previous step and utilizing original data. Consequently, *EEV* shows the maximum cost that we would be prepared to pay to disregard uncertainty. In this paper, the *VSS* has a positive value that demonstrates the advantage of two-stage stochastic programming.

Reducing the relief time in DRP is critical in minimizing damages and losses. Experts in the area of disaster management try to plan the best use of facilities in minimizing the

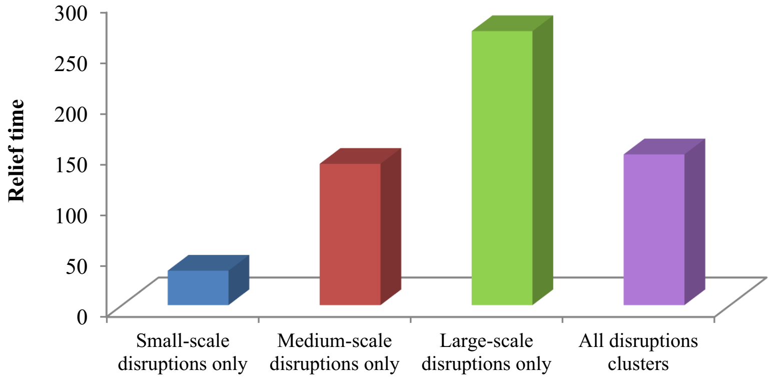

Figure 6.
Total relief operation time for different disruption clusters

response time for helping the affected people. One of the influential factors on relief time is the maximum number of DCs that can be opened. Furthermore, selecting vehicles with appropriate capacities for the distribution of items affects the duration of relief operations. Hence, the simultaneous effects of these two parameters – number of DCs and capacity of vehicles – on total relief time is analyzed in this study.

Figure 7 shows the percentage decrease in the total relief time through increasing the capacity of the vehicles and maximizing the number of DCs that can be opened compared to the current situation. It should be noted that the maximum changes in parameters have been estimated by disaster management experts, based on the conditions and traffic flow of the routes.

According to Figure 7, a general observation illustrates that the total relief time decreases as the capacity of the vehicles and maximizing the number of DCs increases. Furthermore, for any capacity of the vehicles, increasing the maximum number of available DCs decreases the relief time since vehicles would start their operation from DCs closer to the AAs, which would consequently decrease the travel distance and the relief time. As an example, a 15% increase in the capacity of relief vehicles with a maximum of 2 DCs results in a 14% decrease in total relief time. However, having 3 or 4 DCs would provide 18% and 19% decreases in the relief time, respectively. Therefore, it is with no doubt that for any capacity of the vehicles, having 3 or 4 DCs rather than 2 would

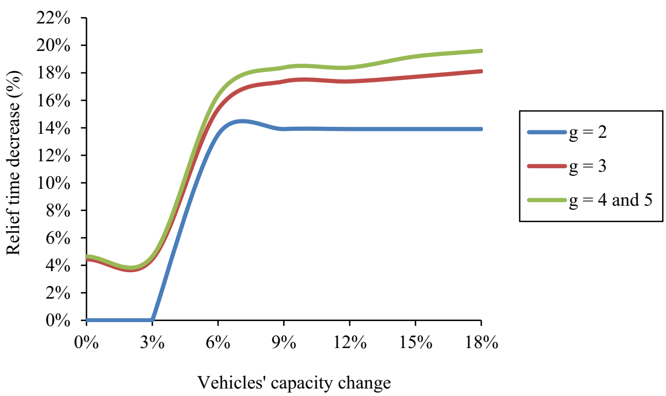

Figure 7.
Effect of changing the number of DCs and capacity of vehicles on relief time

provide a minimum of 1% and 2%, and a maximum of 4% and 6% decreases in total relief time, respectively. Although these changes may seem insignificant, many people can survive with a slight decrease in the relief times.

Moreover, increasing the capacity of vehicles with a constant maximum number of DCs would likewise decrease the total relief time because each vehicle can load more items and upon leaving a DC, it can serve a larger number of AAs. This would decrease the need for loading and dispatching new vehicles on the same routes, consequently decreasing the total relief time. For instance, with a maximum number of 3 DCs, a 6% and 9% increase in the capacity of fleet of vehicles would result in a 15% and 17% decrease in total relief time, respectively. Also, a 15% increase in vehicle capacity will reduce the total service time by 18%, but this trend becomes stable at a 15% increase of vehicles capacity, and no more decrease is observed in total relief time beyond this point.

Given the aforementioned results, experts in the field of disaster management can select optimal capacity for relief vehicles fleet in order to most efficiently distribute relief items according to the expected service time to evacuees and available budget and facilities to open DCs.

7. Conclusion and future research suggestions

One of the critical challenges for decision-makers in DRP is to provide relief items for evacuees in AAs, of which using efficient routes and scheduling of relief operations are not well considered, and this negligence increases the severity of casualties. Delivering of relief items is carried out using fleets of relief vehicles from multiple DCs to the AAs. Limitations in available resources such as the number and capacity of relief vehicles and lack of accurate information regarding evacuees in each AA impede relief operations. Moreover, upon the occurrence of a disaster, unexpected accidents such as route disruptions (destruction of routes, bridges, and tall buildings) impede relief operations and supply of relief items. To address this problem, designing a proper distribution system by disaster management experts is critical, and requires efficient and prompt decisions to minimize the risk of damages. Therefore, the present study proposed a two-stage stochastic programming model for the distribution of relief items from DCs to AAs, with the objective of minimizing the expected value of arrival time of relief vehicles to AAs. This model would help experts in the field of disaster management to prepare an integrated relief operation schedule and enhance the efficiency of decisions in DRP.

In the proposed model, first-stage decisions included locating DCs, and second-stage decisions included R/S of vehicles to reach different AAs. For the relief operation, the demand for relief items in each AA and the disrupted routes were defined as uncertain parameters. To consider the situations more realistically, it was presumed that the demand of each AA could be fulfilled through multiple vehicles and DCs and vehicles had limited capacity.

To verify the applicability of the proposed model, the model was executed on a case study. By carrying out a simultaneous sensitivity analysis on the capacity of vehicles and the maximum number of DCs that can be opened, optimal values for these parameters were determined, that would help making optimal decisions upon the occurrence of a disaster to decrease total relief time and to maximize the exploitation of available facilities. Suggestions for future studies in this topic are as follows:

- consideration of disruptions in DCs;
- improvement of the present model to a multi-period state, with multiple objectives for distribution of relief items to the AAs in shorter time periods;
- solving the model for a larger-scale problem using heuristic and metaheuristic algorithms.

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